

# CY384: Cybersecurity Lab Work II



## Important Instructions for all students:

Some students may delay or fail to submit their work on time for various reasons. However, please note that any student who does not submit their work by the deadline will not be given a second chance. It is essential to follow all instructions carefully. If you encounter any issues, you must communicate with the instructor at least three days before the deadline.

Additionally, make sure to read all instructions in this document thoroughly.

## Scenario

Stemi Security Solutions is a security firm located in Ghana. With the support of a dedicated team specializing in computer security and forensics, the company has built a strong reputation in security, incident response, and forensic investigations. To harness the unique talents of each member, the company has decided to organize members into three teams.

## Assignment Tasks

Stemi Security Solutions has requested that each member choose the team they wish to join. Each team has been assigned specific tasks that members must complete.

### Team One Task:

Every member of Team One is required to investigate, design, and implement advanced possible solutions on one of the following topics:

1. Trustworthy and reliable agentic AI for security operations
2. Threat, intrusion, and anomaly detection in agentic AI systems and cloud environments
3. Reinforcement learning for information security
4. AI-powered incident response and red-teaming
5. AI for malware analysis and detection, with a focus on cloud environments or devices
6. Securing Automotive Systems
7. Securing Cyber-physical Systems
8. Security for the Metaverse
9. Securing Digital Twins
10. Security in Telepresence
11. Data Privacy and Ethics in Telepresence
12. Security and Trust Management in IoT
13. Security for Smart Grids
14. Access Control in IoT
15. Privacy-Enhancing Technology for Identity Management
16. Passwordless Authentication
17. Threat Modeling, Security and Risk Analysis of Identity Management Solutions
18. Securing Identities for Financial, Governmental and Health Services
19. Mobile and Strong Authentication
20. Non-Human Identities
21. Decentralized Identity and Self Sovereign Identity
22. Ransomware Resilience
23. AI for simulating cyberattack scenarios

24. DoS/DDos reduction via adaptive traffic analysis
25. Security mechanism against critical infrastructure sabotage
26. Lightweight cryptography for constrained IoT devices
27. API security testing, monitoring, and exploitation prevention
28. Forensics in AI-enabled environments
29. Detecting covert communication channels
30. Adversarial ML, model poisoning, dataset integrity monitoring
31. Security in Public Safety Technologies
32. Vehicle forensics

### **Team Two Task:**

Each member of Team Two is to explore the Common Vulnerabilities and Exposures (CVE) library and select three CVEs. These will be used to identify vulnerabilities in a Ghanaian website or application. All members of this team must sign a disclaimer stating that their access to information is solely for academic purposes. All disclaimers are to be submitted at the Instructors office by 12<sup>th</sup> June 2026. Each member will write a proposal detailing the three selected CVEs and the website or application they intend to test. Each member should select unique CVEs, website or application. Members with the same CVEs, website or application will receive a score of 0. Members should refrain from using a student website or application. Scores will be allocated on the applications or websites selected.

### **Team Three Task:**

Members of this team are to write a research paper that addresses the technical aspects of topics related to computer security, cybersecurity, or digital forensics. The content of the paper does not need to be original in terms of scientific findings, but it should provide a unique perspective and clearly demonstrate the author's contribution to the subject. Previously published work will not be accepted. Please visit

[https://docs.google.com/spreadsheets/d/1awrqgJ9a\\_RLi6MKRK2D5VTgl2syRcxt5IF1UHi7IF2k/edit?usp=sharing](https://docs.google.com/spreadsheets/d/1awrqgJ9a_RLi6MKRK2D5VTgl2syRcxt5IF1UHi7IF2k/edit?usp=sharing)

for conferences or journals where your paper can be submitted for publication. The page will be updated periodically. Any member who submits their publication to a journal that is not approved by the instructor will receive a score of zero. Your paper must be accepted for publication on or before September 1, 2026. If a submitted paper is not accepted, the member will not receive any marks.

### **Final Deliverables for teams:**

All members in Teams One and Two must submit a proposal as well as a final report summarizing their work. Additionally, each member will perform a demonstration of their work.

All members in Team Three are to submit a copy of their accepted papers by 1<sup>st</sup> September 2026.

### **Deliverables**

1. Proposal: This should be structured in response to either Teams One or Two tasks outlined above. The proposal must adhere to the university's standards for project writing.
2. Main Report: The report should be organized in response to the assignment task provided above. It must also follow the university's standards for project writing.
3. Artefact: The system you create or exploit, along with all related files (e.g., virtual machines in the cloud, technical notes, etc.), must be made available to the instructor.

\*All deliverables are mandatory for Teams One and Two. Reports that are written or generated using ChatGPT or any AI tool will receive a score of 0.

### **Deadlines (Teams One &Two)**

|                                           |                                                |     |
|-------------------------------------------|------------------------------------------------|-----|
| 1. Disclaimer Submission (Team Two):      | 12 <sup>th</sup> June 2026                     |     |
| 2. Proposal Submission:                   | 14 <sup>th</sup> June 2026                     | 20% |
| 3. Proposal Justification (Presentation): | 15-19 <sup>th</sup> June 2026                  | 10% |
| 4. Initial Implementation (Presentation): | 13 <sup>th</sup> -17 <sup>th</sup> July 2026   | 10% |
| 5. Final Demonstration (Presentation):    | 24 <sup>th</sup> -28 <sup>th</sup> August 2026 | 20% |
| 6. Main Report Submission:                | 30 <sup>th</sup> August 2026                   | 40% |

### **Deadline (Team Three)**

|                                        |                                |
|----------------------------------------|--------------------------------|
| 1. Accepted Paper Details Submission : | 1 <sup>st</sup> September 2026 |
|----------------------------------------|--------------------------------|

**All deliverables must be electronically submitted using VLE by 11:59 pm on the due date.**

Please leave sufficient time to meet this deadline and do not leave the handing-in of assignments to the last minute. You need to allow plenty of time for system problems or other issues that may arise.